Do Sparse Autoencoders Preserve What a Language Model Uses?

A Layer-wise Audit of Reconstruction, Information, Behavioural and Causal Faithfulness in GPT-2 Small

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Abstract

Sparse autoencoders (SAEs) are widely used to explain what large language models represent, which presumes that an SAE is a faithful stand-in for the activations it replaces. We audit that presumption at all twelve layers of GPT-2 Small for three SAE architectures (ReLU, TopK and a sigmoid-surrogate JumpReLU; 108 SAEs at matched sparsity, three seeds each), separating four levels of faithfulness: reconstruction, task-information preservation, behavioural faithfulness and causal faithfulness. Reconstruction degrades with depth (OWT test explained variance 0.93–0.95 at layers 0–2, 0.76–0.82 at layer 11). Within a layer, reconstruction error predicts how much splicing the SAE damages the model's predictions, but not how much truth information is lost. The apparent collapse of a frozen truth probe on SAE reconstructions is largely reproduced by random noise of the same norm; re-trained readers show a smaller but real loss, and at equal reconstruction error the SAEs never preserve task information better than PCA. Using an equal-length persona-instruction task built to remove a length shortcut, we locate where GPT-2 carries the instruction's effect and decompose it: the SAE-reconstructed activation difference restores 93–97 % of the effect at the layer-1 persona token but only 62–68 % at the layer-8 and layer-11 final token, with the remainder in the reconstruction error or in an interaction between reconstruction and error. Short fine-tuning at layer 6 improves task reconstruction and, with an output-KL term, behavioural faithfulness for TopK, but no method meets its pre-registered criterion for truth preservation. All results except the reconstruction table are validation-stage; a single pre-registered held-out evaluation is pending.

PENDING: Abstract to be revised after the multi-layer improvement experiment (IMP-L), E1–E3 and the final held-out evaluation (M5). Numbers above are validation-stage unless marked TEST.

1 Introduction

Sparse autoencoders decompose a model's dense activations into a larger set of sparsely active latents, many of which look human-interpretable [3, 4]. They have become a standard tool of mechanistic interpretability, with architectures that trade off reconstruction against sparsity in different ways [5, 13]. Interpretability claims made through an SAE — that the model represents truth with certain features, or that a feature controls a behaviour — inherit an assumption: that the SAE is a *faithful* replacement for the activation it reconstructs.

Faithfulness has several distinct meanings that are often conflated:

1. **Reconstruction**: the reconstruction $\hat{x}$ is close to the activation x.

2. **Information preservation**: task information that the model encodes (here, whether a statement is true; whether a persona instruction is honest) can still be read from $\hat{x}$.

3. **Behavioural faithfulness**: the model behaves the same when $\hat{x}$ replaces x.

4. **Causal faithfulness**: $\hat{x}$ carries the specific causal effect the model uses for a task.

Explained variance is dominated by high-variance directions shared by all inputs, while task-relevant differences can be small, low-variance directions. Good reconstruction therefore does not guarantee the other three properties. This paper measures each level separately, at every layer of GPT-2 Small [12].

Contributions.

- A matched-sparsity comparison of ReLU, TopK and JumpReLU SAEs at all 12 layers of GPT-2 Small, with three seeds (§6).
- A four-level information test that separates readout shift from information loss using frozen and re-trained readers, norm-matched noise and PCA references (§7).
- A within-layer analysis showing that reconstruction error predicts behavioural damage but not truth readout, and that earlier pooled correlations were artefacts of mixing layers (§8).
- A shortcut-free persona-instruction task, a pre-registered causal site sweep, and a decomposition of the causal effect into reconstruction, error and interaction parts (§9).
- Pre-registered fine-tuning experiments, including an audit that found our own earlier truth-aware objective to be circular (§10).

Scope of the claims. All in-house SAEs have 3072 latents (4× expansion), target L0 = 128 and are trained on 101,600 OpenWebText tokens. The results describe these SAEs; whether they hold for larger, longer-trained SAEs is tested only partially (§13).

PENDING: Final framing paragraph (Review #8: lead with the audit's negative finding) after IMP-L and M5.

2 Related Work

Sparse dictionary learning for interpretability. Bricken et al. [3] and Cunningham et al. [4] showed that SAEs trained on transformer activations yield many interpretable latents. Gao et al. [5] introduced TopK SAEs and studied scaling; Rajamanoharan et al. [13] introduced JumpReLU SAEs with a learned threshold and a straight-through estimator. Our JumpReLU uses a simpler sigmoid surrogate and is not the published estimator (§5). Public SAE suites exist for GPT-2 Small via SAELens [1] and for Gemma 2 via Gemma Scope [8].

Evaluating SAEs beyond reconstruction. SAEBench [7] evaluates SAEs on a broad set of downstream metrics and finds that proxy metrics such as reconstruction and sparsity do not reliably predict practical usefulness. AxBench [14] finds that simple baselines outperform SAE-based methods for concept detection and steering. Paulo et al. [11] automate the interpretation of SAE features at scale. Our audit is narrower but more controlled: one model, one task family per question, and pre-registered comparisons with explicit noise, PCA and synonym controls.

Training for downstream faithfulness. Braun et al. [2] train SAEs end-to-end with a KL objective on the model's output. Our output-KL fine-tuning (§10) is inspired by this but is not a reproduction: it fine-tunes existing local SAEs for 500 steps and keeps the local reconstruction term.

Truth and honesty representations. Marks and Tegmark [9] find linear truth structure in LLM activations (the Geometry of Truth datasets we use); Zou et al. [15] introduce representation engineering with persona-contrast prompts, from which our original RepE task was derived.

PENDING: Expand to the 15-reference review from PR #16 (Review #8 H3b); add a differentiation statement per work. Every citation must be checked against the original paper before submission.

3 Background: Transformer Activations and SAEs

GPT-2 Small has 12 transformer blocks with a residual stream of width $d = 768$. We study the activation entering block L, `blocks.`L`.hook_resid_pre`, for $L = 0, \ldots, 11$; at $L = 0$ this is the token plus position embedding.

An SAE with m latents maps $x \in \mathbb{R}^d$ to

$$z = \sigma\big(W_e(x - b_d) + b_e\big) \in \mathbb{R}^m, \qquad \hat{x} = W_d z + b_d, \qquad e = x - \hat{x}, \tag{1}$$

with decoder columns renormalised to unit norm after every optimisation step. The three activation functions are

$$\text{ReLU:} \quad z = \mathrm{ReLU}(p), \qquad \text{penalty } \textstyle\sum_i z_i, \tag{2}$$

$$\text{TopK:} \quad z = \mathrm{TopK}_k\big(\mathrm{ReLU}(p)\big),\ k = 128, \qquad \text{no penalty}, \tag{3}$$

$$\text{JumpReLU (surrogate):} \quad z = \mathrm{ReLU}(p) \odot H(p - \theta), \qquad \text{penalty } \textstyle\sum_i \sigma\big((p_i - \theta_i)/T\big), \tag{4}$$

where p is the pre-activation, H the step function and θ a learned per-latent threshold whose gradient uses a sigmoid straight-through estimator with $T = 0.1$. The training loss is $\mathrm{MSE}(\hat{x}, x) + c \cdot \text{penalty}$, and a multiplicative controller adjusts c every 25 steps towards a target L0 of 128.

Metrics. Explained variance is $\mathrm{EV} = 1 - \sum \|x - \hat{x}\|^2 / \sum \|x - \bar{x}\|^2$ and $\mathrm{NMSE} = 1 - \mathrm{EV}$; L0 is the mean number of active latents. An AUROC gap is $\mathrm{AUROC}(x) - \mathrm{AUROC}(\hat{x})$ for a given reader. Spliced ΔCE is the increase in next-token cross-entropy when $\hat{x}$ replaces x at every non-BOS position; loss recovered is $(\mathrm{CE}_{\text{mean-abl}} - \mathrm{CE}_{\mathrm{SAE}})/(\mathrm{CE}_{\text{mean-abl}} - \mathrm{CE}_{\mathrm{orig}})$. Causal quantities are defined in §9.

4 Research Questions and Hypotheses

Main question. Can SAE reconstruction of frozen LLM activations be improved while preserving truth-related information and causal behaviour across transformer layers?

SQ1 (reconstruction). How do ReLU, TopK and JumpReLU SAEs compare across the 12 layers at matched sparsity?

SQ2 (information). Which task information survives reconstruction, and is the loss specific to SAEs compared with generic perturbations of the same size and with PCA at the same reconstruction error?

SQ3 (behaviour). How much does splicing the SAE in damage GPT-2's predictions, and does reconstruction error predict that damage between SAEs at the same layer?

SQ4 (causal). Where does GPT-2 carry a persona-instruction effect, and how much of it survives in the SAE reconstruction?

SQ5 (improvement). Can fine-tuning improve faithfulness at matched sparsity, and at which layers?

Evidential status. Each result is labelled TEST (held-out split, evaluated once under a written protocol), VALIDATION, or PRELIMINARY (a validation pilot whose pre-registered criterion was not met or which is unreplicated). Every experiment from M0 onwards was pre-registered with file hashes before it ran.

The project's original proposal asked whether SAE features are causally faithful to their auto-generated labels. The scope was narrowed to the questions above after instructor reviews and internal audits (Appendix A).

5 Datasets and Experimental Setup

OpenWebText (OWT). 1000 documents sampled from `Skylion007/openwebtext` (streaming shuffle, seed 42, 100–20,000 characters, exact deduplication), split 800/100/100 by document. Each document is truncated to 128 tokens; excluding BOS this gives 101,600 / 12,700 / 12,700 training, validation and test tokens.

Geometry of Truth (GoT). The `cities` and `neg_cities` sets [9] (1496 statements each) form the "geographic" truth task, with city-grouped splits of 1046/224/226 per set. The numerical sets have a surface shortcut (the second number alone gives AUROC 0.84) and are used only as a diagnostic.

RepE-LM persona task. Our original persona prompts ("Pretend you're an honest/untruthful person") had a length shortcut: *untruthful* is three tokens longer than *honest*, and prompt length alone predicts the label with AUROC 0.80. We therefore built RepE-LM: 506 statements (train 417, validation 89) times 20 prompt variants, in which the two prompts of each pair differ in exactly one token at a fixed position. Templates T1–T4 swap truthful/deceptive, trustworthy/dishonest, sincere/lying and facts/lies, each with an honest-synonym control; T5–T6 are held out; N1–N2 swap neutral words (tall/short, red/blue). Seven pre-registered gates (Table 1) verify the construction.

SAE training. For each layer, architecture and seed (42, 1042, 2042) we train one SAE with width 3072, Adam (learning rate 3×10^{-4}), batch 256 sampled with replacement, gradient clipping at 10, and one uninterrupted run of 15,000 steps (3.84M samples, about 38 passes over the training tokens). Architectures are compared only when every run's L0 is within $\pm 10\,\%$ of the target; 107 of 108 runs qualify (layer-0 JumpReLU seed 42 does not).

Splits. Probes and selections use training data; results are reported on validation unless labelled TEST. The RepE-LM test split has never been built or accessed. The GoT test split was scored in earlier frozen-probe evaluations and is treated as previously exposed (Appendix B).

Table 1: M1 RepE-LM construction and gates (pre-registered pass rules). Source: `results/repe/repe_lm_gates/gates.json`.

Gate	Status	Evidence (train statements)
K1 construction	PASS	6,072 pairs; max residual difference before the swapped token 0.0 at all 12 layers
K2 shortcuts	PASS	max deviation of any shortcut AUROC from 0.5: 5.6e-17
K3 positive control	PASS	unmasked bag-of-words AUROC 1.000
K5 persona meaning	PASS	leave-one-pair-out AUROC 0.701 at layer 8 (CI low 0.689); passing layers [8, 11]
K6 leakage	PASS	0 exact overlaps; 0 same-label near-duplicates
K7 behaviour	PASS	median KL(H‖U) 0.0186 vs KL(H‖H′) 0.0092 nats; ratio 1.99; Wilcoxon $p = 3e-68$

Table 2: Reconstruction on the OWT test split (TEST; 12,700 tokens), width 3072, target L0 = 128. Layer-0 JumpReLU includes the unmatched seed 42 (L0 73.6). Source: `results/openwebtext/reconstruction_evaluation/uninterrupted_protocol/all_layers_summary.csv`.

Layer	OWT test EV (mean±sd, 3 seeds)			OWT test L0		
	ReLU	TopK	JumpReLU	ReLU	TopK	JumpReLU
0	0.950±0.000	0.911±0.000	0.826±0.101	135.1	125.3	110.3
1	0.937±0.000	0.921±0.001	0.933±0.000	132.2	128.0	129.3
2	0.936±0.000	0.926±0.000	0.935±0.002	131.9	128.0	129.0
3	0.905±0.000	0.906±0.000	0.915±0.001	131.7	128.0	129.1
4	0.872±0.000	0.884±0.000	0.893±0.001	131.1	128.0	129.1
5	0.848±0.000	0.868±0.000	0.875±0.000	130.8	128.0	129.0
6	0.832±0.001	0.857±0.000	0.865±0.000	131.0	128.0	129.1
7	0.819±0.001	0.850±0.000	0.855±0.000	131.4	128.0	129.9
8	0.805±0.000	0.840±0.000	0.844±0.000	131.9	128.0	130.3
9	0.789±0.001	0.831±0.000	0.833±0.000	132.4	128.0	131.0
10	0.773±0.000	0.822±0.000	0.823±0.000	132.7	128.0	131.6
11	0.757±0.001	0.815±0.000	0.815±0.000	133.2	128.0	132.1

6 Reconstruction and Sparsity

Table 2 and Figure 1 (left) give OWT test reconstruction for all 108 SAEs (TEST). Explained variance falls with depth for every architecture, from 0.93–0.95 at layers 0–2 to 0.757 (ReLU) and 0.815 (TopK, JumpReLU) at layer 11. Among matched runs, ReLU leads at layer 1, ReLU and JumpReLU tie at layer 2, JumpReLU leads at layers 3–9, and JumpReLU and TopK tie at layers 10–11. Seed standard deviations are ≤ 0.001 except for layer-0 JumpReLU, whose seed 42 is unmatched (L0 73.6; about 96 % of latents dead) and excluded from rankings.

Sparsity is matched on OWT only. ReLU runs 2–6 % above the target L0. A within-layer adjustment using the L0 elasticity measured in an earlier sweep changes no ranking except two near-ties (M0). The architectures differ strongly in effective width: of 3072 latents, ReLU keeps 2920–3063 alive, TopK 1711–2851 and JumpReLU 469–2947. On task prompts the number of active latents is not matched (on GoT final tokens, TopK uses 128, ReLU 32–110 and JumpReLU 23–92 at layers 1–11), which confounds architecture comparisons on task information (§7).

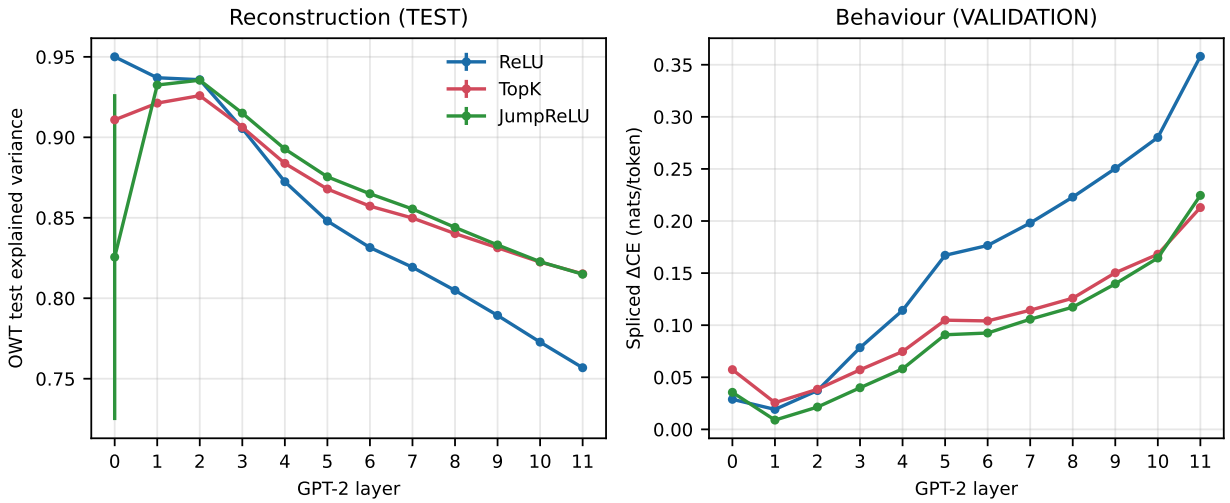


Figure 1: Left: OWT test explained variance by layer (TEST; error bars: seed sd). Right: spliced next-token cross-entropy increase (VALIDATION). Sources: Tables 2 and 4.

PENDING: E1 (TopK trained at several k to match task-level L0) and E2 (public `gpt2-small-res-jb` reference; not matched in width, data or L0).

7 Information Preservation

Frozen probes overstate the loss. A logistic truth probe fitted on original activations and applied unchanged to SAE reconstructions loses 0.28–0.29 AUROC on GoT (Table 3, Figure 2). Earlier versions of this project read this as truth information being lost. Adding isotropic noise with the same norm as the SAE error causes almost the same drop (0.25–0.26), and a simpler mass-mean reader loses only 0.02–0.03. Most of the frozen-probe collapse is therefore sensitivity of that reader to any perturbation of this size, not SAE-specific loss.

A smaller, real loss remains. Readers re-trained on the reconstructions are 0.07–0.12 AUROC worse than readers trained on the original activations for GoT, and 0.14–0.18 worse for the persona task. Re-training never recovers the original performance, so no cell is classified as "readout shifted"; no cell is "preserved". The task label can be decoded from the reconstruction error alone (AUROC 0.62–0.77), so the lost information is still in the activation, outside the SAE's reconstruction.

SAEs are never better than PCA. At the same OWT reconstruction error, PCA keeps at least as much task information as the SAE in every cell; the SAE is significantly worse in 52 of 81 GoT cells, 59 of 99 persona cells and 81 of 90 neutral cells (one-sided paired bootstrap, BH-corrected).

Persona versus generic change. Part of the persona loss is generic one-word-change information: the re-trained gap is 0.07–0.13 larger for persona than for neutral swaps at layers 3–8, and about equal at layers 10–11.

Table 3: M2 information preservation (VALIDATION): AUROC gaps (original − reconstructed; ↓), mean over informative layers (GoT 4–11, RepE-LM 1–11) and 3 seeds. "Noise adds isotropic noise with the norm of the SAE error; "PCA is matched to the SAE's OWT validation NMSE. Error AUROC: label decoded from the error term alone. Source: `results/shared/m2_four_level_faithfulness/summary_by_layer_arch.csv`.

Task	SAE	Frozen log.	Noise	Frozen m.-mean	Re-trained log.	PCA re-tr.	MLP	Error AUROC
GoT	ReLU	0.287	0.259	0.033	0.066	0.042	0.049	0.675
GoT	TopK	0.275	0.246	0.024	0.075	0.032	0.037	0.615
GoT	JumpReLU	0.286	0.257	0.026	0.118	0.030	0.066	0.632
Persona	ReLU	0.198	0.076	0.091	0.179	0.131	0.232	0.773
Persona	TopK	0.162	0.072	0.066	0.147	0.116	0.190	0.719
Persona	JumpReLU	0.155	0.064	0.062	0.143	0.114	0.177	0.721
Neutral	ReLU	0.113	0.048	0.049	0.099	0.066	0.033	0.636
Neutral	TopK	0.098	0.046	0.038	0.090	0.064	0.031	0.615
Neutral	JumpReLU	0.094	0.043	0.035	0.083	0.062	0.031	0.619

Architecture differences are confounded. On truth, ReLU loses least and JumpReLU most, the reverse of the reconstruction ranking; on persona, ReLU loses most. Because task-level L0 is not matched, we do not interpret these differences as architecture effects until E1 is run.

PENDING: M5 held-out confirmation; E1 task-level sparsity control; E2 external SAE reference.

8 Behavioural Faithfulness

Splicing the reconstruction into GPT-2 at every non-BOS position increases next-token cross-entropy by 0.009–0.026 nats at layer 1 and by 0.213–0.358 nats at layer 11 (Table 4, VALIDATION); loss recovered relative to mean ablation stays at or above 0.93. An identity splice changes nothing (exactly 0). The true-versus-false likelihood preference on GoT `cities` (0.82 originally) is unchanged within ±0.025 through layer 8 and drops by 0.04–0.09 at layers 9–11.

SAE error is more harmful than random error. Replacing the SAE error by isotropic noise of the same norm gives a smaller cross-entropy increase in most runs: the ratio of SAE-induced to noise-induced damage is 0.76–2.89 per run (median 1.64); 7 of 107 runs are below 1, all at layers 1–3.

Within a layer, reconstruction error predicts behavioural damage but not truth readout (M0). Table 5 reports within-layer regressions over the 107 matched SAEs. Lower OWT NMSE goes with less splice damage ($\beta = 0.165$ per unit log NMSE, Holm $p = 0.0008$; per-layer Spearman ρ 0.80–0.97 from layer 3). For the frozen GoT truth gap, the formal test passes (Holm $p = 0.044$), but architecture explains 99.8 % of the within-layer NMSE variance and the association disappears without architecture fixed effects ($p = 0.89$); GoT reconstruction error has no association. Two earlier claims are corrected: a pooled correlation of 0.89 between reconstruction and persona preservation is 0.05 within layer, and the range of the SAE-versus-noise damage ratio is the one given above.

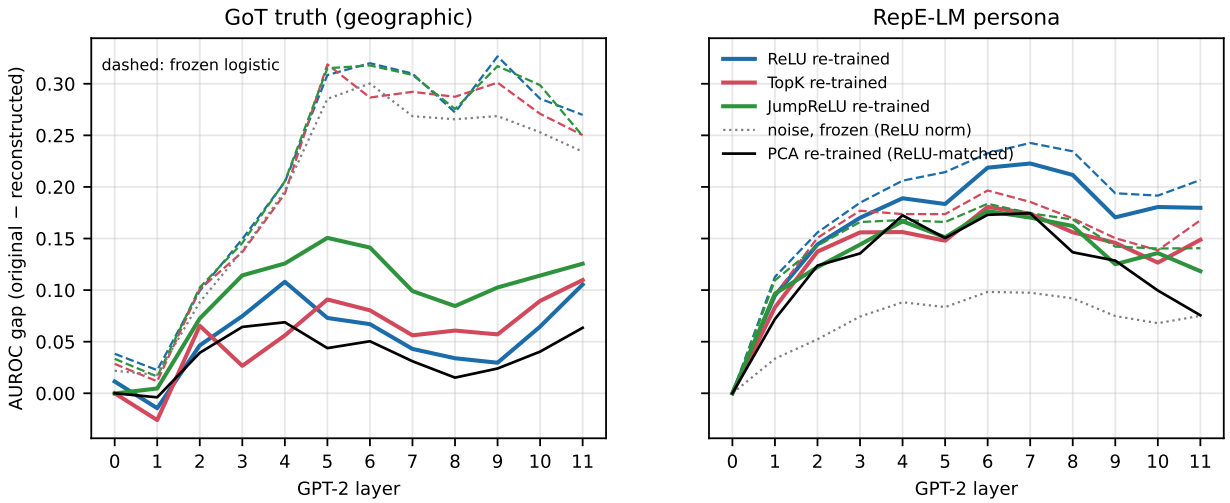


Figure 2: Information preservation by layer (VALIDATION). Dashed: frozen logistic gap; solid: re-trained logistic gap; dotted grey: frozen gap under norm-matched noise; black: PCA at the ReLU SAE's error.

9 Causal Faithfulness

Where GPT-2 carries the persona effect (M3). For each RepE-LM pair we run GPT-2 on the honest (H) and the dishonest (U) prompt and measure $\mathrm{KL}(p_H \| p_U)$ on the next token (mean 0.033 nats). Copying the honest activation into the dishonest run at one layer and position group restores a share E of that divergence. Figure 3 shows that the effect is carried by the persona word early (layer 1: $E = 0.97$) and by the final token late (layer 11: $E = 0.95$), with the handover at layers 7–8; the instruction words after the persona slot carry up to 0.28. Of 65 cells, 50 pass the pre-registered test (Holm-corrected, above the 95th percentile of 200 matched-norm random vectors). The pre-registered rule selected the layer-1 persona slot and the layer-11 final token, both confirmed on validation; a pre-registered amendment added the layer-8 final token (train $E = 0.505$), confirmed without SAEs. A same-meaning synonym swap travels the same route at about half the size. Template T1 (truthful/deceptive) has no effect beyond its synonym control and serves as a control case; T2–T4 are the primary set.

How much of the effect the SAE carries (M4). At a site, let $\Delta = x_H - x_U$, $\Delta_{\mathrm{rec}} = \hat{x}_H - \hat{x}_U$ and $\Delta_{\mathrm{err}} = e_H - e_U$, so $\Delta = \Delta_{\mathrm{rec}} + \Delta_{\mathrm{err}}$ exactly. With $T(v) = \mathrm{KL}(p_H \| p_U) - \mathrm{KL}(p_H \| p_{U+v})$, we define $T_{\mathrm{full}} = T(\Delta)$, $T_{\mathrm{rec}} = T(\Delta_{\mathrm{rec}})$, $T_{\mathrm{err}} = T(\Delta_{\mathrm{err}})$, the interaction $I = T_{\mathrm{full}} - T_{\mathrm{rec}} - T_{\mathrm{err}}$, and Shapley shares $\phi_{\mathrm{rec}} = (T_{\mathrm{rec}} + T_{\mathrm{full}} - T_{\mathrm{err}})/2$. The patch adds v to the dishonest run's original activation; it is not a full splice. All exact checks pass (Δ identity error 9.5×10^{-7}; T_{full} reproduces M3 to 7×10^{-8}).

Table 6 and Figure 4 show that the reconstructed difference restores 93–97 % of the effect at the layer-1 persona token, but only 62–68 % at the layer-8 and layer-11 final token. There, ReLU's error part alone carries 40–45 %; for TopK and JumpReLU, the interaction is large ($|I|/T_{\mathrm{full}}$ 0.26–0.38), so their shares cannot be read additively. The persona-minus-synonym difference is preserved in the reconstruction at every site. Higher explained variance does not imply higher causal preservation: at layer 11, TopK has higher EV than ReLU (0.815 vs 0.757) but keeps less of the effect (0.62 vs 0.68).

Table 4: Behavioural faithfulness: GPT-2 with the SAE reconstruction spliced in at every non-BOS position (OWT validation; original CE 3.396 nats/token; matched seeds only). Source: `results/openwebtext/spliced_evaluation/summary.csv`.

Layer	Spliced ΔCE (nats/token, ↓)			Loss recovered (↑)		
	ReLU	TopK	JumpReLU	ReLU	TopK	JumpReLU
0	0.029	0.057	0.036	0.996	0.991	0.995
1	0.019	0.026	0.009	0.998	0.997	0.999
2	0.037	0.038	0.021	0.995	0.994	0.997
3	0.078	0.057	0.040	0.987	0.991	0.994
4	0.114	0.075	0.058	0.984	0.990	0.992
5	0.167	0.105	0.091	0.967	0.980	0.982
6	0.177	0.104	0.092	0.966	0.980	0.982
7	0.198	0.114	0.106	0.963	0.979	0.980
8	0.223	0.126	0.117	0.958	0.976	0.978
9	0.250	0.150	0.140	0.951	0.971	0.973
10	0.280	0.168	0.164	0.947	0.968	0.969
11	0.358	0.213	0.225	0.930	0.958	0.956

Table 5: M0 within-layer regressions (VALIDATION): outcome ∼ β·predictor + layer FE + architecture FE + log L0, Freedman–Lane permutation within layer. The GoT row passes formally but vanishes without architecture fixed effects ($p = 0.89$; see text). Source: `results/shared/m0_within_layer_reanalysis/within_layer_regressions.csv`.

Predictor → outcome	β	95% CI	p_{perm}	p_{Holm}	n
log OWT NMSE → GoT frozen-logistic gap	0.268	[0.056, 0.477]	0.0146	0.0438	72
log GoT relative error → GoT frozen-logistic gap	0.005	[-0.059, 0.080]	0.8680	1.0000	72
log OWT NMSE → OWT spliced ΔCE	0.165	[0.134, 0.265]	0.0002	0.0008	107
log GoT relative error → OWT spliced ΔCE	-0.012	[-0.060, 0.028]	0.6267	1.0000	107

SAE-in-the-loop feature patches. Encoding both prompts, setting the dishonest prompt's top-k features to their honest values, decoding and keeping the error term restores more than 200 frequency- and magnitude-matched random feature sets in all 27 SAE × site combinations ($p = 0.005$, the minimum attainable). The effect is distributed: one feature carries 2–10 % of T_{rec} and 64 features 47–74 %. This shows that SAE-defined directions are causally sufficient when inserted; it does not show that GPT-2 uses SAE features.

PENDING: Minimum detectable effect statement (Review #8 critique 3); E3 causal preservation of fine-tuned layer-8 SAEs; M5 held-out confirmation.

10 Fine-tuning Experiments

All fine-tuning starts from a trained SAE and runs 500 Adam steps (learning rate 10^{-5}) with the sparsity controller continued from the base coefficient; ReLU and JumpReLU must stay within ±5 % of their base L0. Truth is measured with probes fitted on original activations of GoT training groups B, which no training objective sees.

A circular objective, found and corrected. A truth-aware objective that distilled a linear probe's logits appeared to remove most of the truth gap. An audit found that the evaluation probe

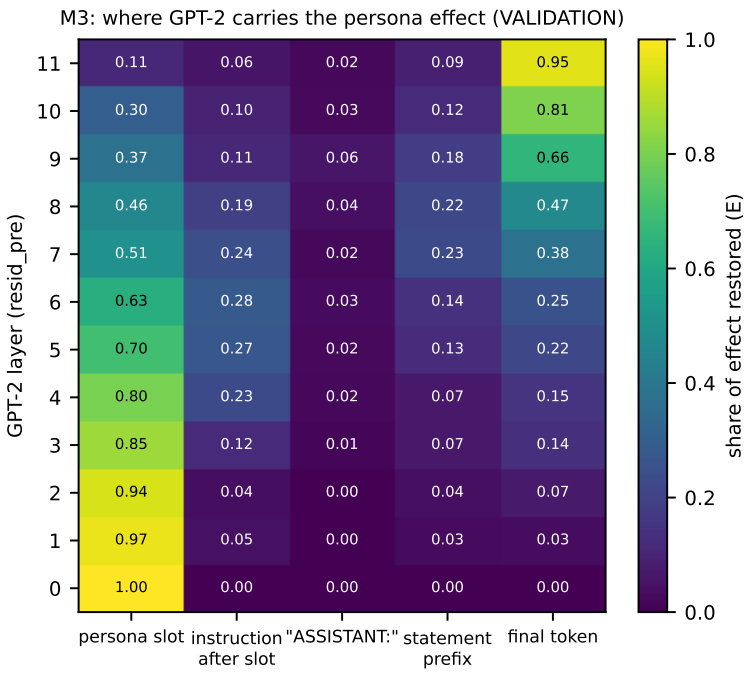


Figure 3: M3 (VALIDATION): share of the persona-instruction effect restored by patching one position group at one layer of GPT-2 Small.

was the training teacher (same rows, labels and weights), so the result was circular. In a corrected design (teacher fitted on groups A, probes on B, random-teacher control) the truth-aware objective is worse than plain reconstruction fine-tuning on every seed (TopK B-MLP gap 0.189 vs 0.107; random teacher 0.165).

Method A and output-KL at layer 6. Table 7 reports Method A (GoT reconstruction with OWT replay) and the output-KL variant. Method A cuts GoT reconstruction error by 72–82 % and lowers the independent MLP truth gap on 18 of 18 paired comparisons against both the base and an OWT-only control, but never by more than 2 sd, the pre-registered criterion (PRELIMINARY). Adding output KL lowers OWT spliced KL (0.115→0.102) and GoT statement KL (0.110→0.038) for TopK, meeting its behavioural criterion without changing the truth gap; for ReLU and JumpReLU the KL term pushes L0 5–11 % above the base, so their gains are not creditable. A KL-weight ablation found no truth trade-off to optimise (INCONCLUSIVE).

10.1 Multi-layer improvement experiment (IMP-L)

PENDING: Layers 3, 8 and 9; base vs OWT-only control vs Method A vs output-KL; 3 architectures × 3 seeds; 95 % t-intervals over seeds; L0 guards. Pre-registered in `results/geometry_of_truth/multilayer_finetune/preregistration.json`. Table generated from `improvement_table.csv` once the experiment has run.

10.2 Causal preservation after fine-tuning (E3)

PENDING: M4 decomposition at the layer-8 final token for the IMP-L layer-8 SAEs.

Table 6: M4 causal decomposition (VALIDATION; templates T2–T4; mean±sd over 3 seeds). Shapley shares in parentheses are not interpretable as additive (pre-registered rule: interaction too large). Source: `results/repe/sae_patch_decomposition/architecture_summary.csv`.

Site	SAE	T_{full} (nats)	$T_{\text{rec}}/T_{\text{full}}$	$T_{\text{err}}/T_{\text{full}}$	$\lvert I\rvert/T_{\text{full}}$	ϕ_{rec}
L1 persona slot	ReLU	0.0435	0.938±0.019	0.14	0.12	0.90
	TopK	0.0435	0.931±0.014	0.16	0.14	0.89
	JumpReLU	0.0435	0.968±0.011	0.03	0.10	0.97
L8 final token	ReLU	0.0220	0.621±0.015	0.45	0.12	0.59
	TopK	0.0220	0.625±0.039	0.09	0.32	(0.77)
	JumpReLU	0.0220	0.665±0.011	0.11	0.26	(0.78)
L11 final token	ReLU	0.0430	0.677±0.014	0.40	0.16	0.64
	TopK	0.0430	0.620±0.004	0.06	0.38	(0.78)
	JumpReLU	0.0430	0.678±0.007	0.11	0.28	(0.78)

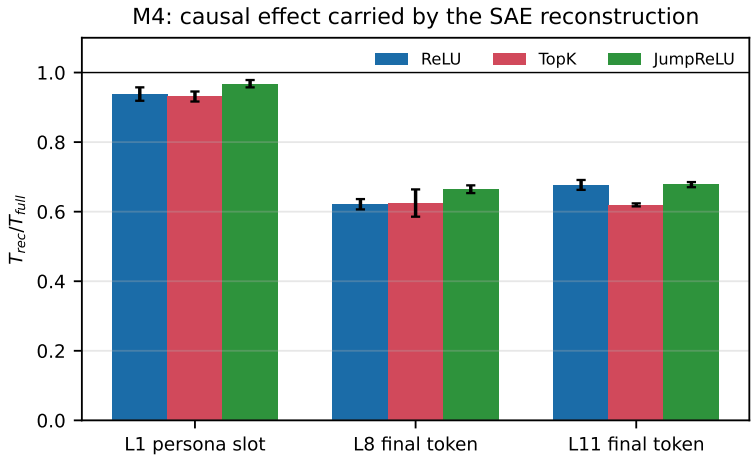


Figure 4: M4 (VALIDATION): share of the causal effect carried by the SAE-reconstructed activation difference ($T_{\text{rec}}/T_{\text{full}}$; mean±sd over 3 seeds; T2–T4).

11 Statistical Analysis

- **Pre-registration.** Every experiment from M0 onwards wrote its hypotheses, criteria and script hashes to `preregistration.json` before running; scripts refuse to overwrite outputs.
- **Within-layer inference (M0).** Fixed-effects regressions with Freedman–Lane permutation within layer (5000 permutations), bootstrap CIs (2000) and Holm correction over four primary pairs.
- **Information (M2).** Statement-grouped bootstrap (paired over the same rows), one-sided tests against noise and PCA references, Benjamini–Hochberg correction across SAEs per task; a 0.02 AUROC tolerance defines "preserved".
- **Causal (M3/M4).** Holm-corrected tests over 65 cells; 200 matched-norm random vectors per cell; synonym and neutral controls; for features, 200 random feature sets matched on activation frequency and $|\Delta z|$.

Table 7: Layer-6 fine-tuning (VALIDATION; mean over 3 seeds). Method A lowered the B-MLP gap on 18/18 paired comparisons but never by more than 2 sd (pre-registered criterion not met). Output-KL rows for ReLU and JumpReLU fail the ±5% L0 guard and are not creditable. Source: `results/geometry_of_truth/method_a_layer6/summary.csv`; `results/geometry_of_truth/e2e_inspired_layer6/summary.csv`.

SAE	Variant	GoT B-MLP gap ↓	GoT rel. error ↓	OWT EV ↑	OWT ΔCE ↓	OWT L0
ReLU	Base	0.321±0.022	0.059	0.828	0.177	134
	OWT-only control	0.319±0.022	0.058	0.829	0.175	134
	Method A	0.245±0.053	0.017	0.828	0.175	134
	Output-KL (B)	0.280±0.020	0.024	0.830	0.145	141
TopK	Base	0.203±0.030	0.034	0.853	0.104	128
	OWT-only control	0.206±0.035	0.033	0.855	0.102	128
	Method A	0.107±0.020	0.006	0.854	0.102	128
	Output-KL (B)	0.114±0.022	0.009	0.852	0.091	128
JumpReLU	Base	0.292±0.026	0.052	0.862	0.092	131
	OWT-only control	0.287±0.027	0.052	0.864	0.091	131
	Method A	0.238±0.056	0.015	0.863	0.091	131
	Output-KL (B)	0.281±0.046	0.022	0.865	0.076	140

- **Seeds.** Three training seeds share the same training data, so seed spread understates uncertainty; comparisons that share seeds are paired. The fine-tuning criterion "consistent" requires the improvement on all three seeds and a mean paired difference larger than twice its standard deviation.
- **Splits.** Fitting and selection use training data; validation results are reported as VALIDATION. Only the reconstruction table is TEST.

PENDING: M5: one pre-registered held-out pass (RepE-LM test, OWT test; GoT test labelled previously exposed), hierarchical bootstrap over statements and seeds, Holm within confirmatory families.

12 Discussion

Reconstruction is a proxy for behaviour, not for information. Within a layer, an SAE with lower reconstruction error damages the model's predictions less, but it does not keep more truth information. Reconstruction metrics average over all directions weighted by their variance; the information that distinguishes true from false statements, or honest from dishonest instructions, lies in small differences between nearly identical activations.

Why the causal effect leaks. M4 makes this concrete. An SAE can reconstruct the honest and the dishonest activation well individually and still reconstruct their difference poorly, because the two reconstruction errors do not cancel. At mid and late sites about a third of the effect is outside the reconstructed difference, and for TopK and JumpReLU part of it exists only when reconstruction and error act together through GPT-2's nonlinear downstream computation.

The negative result is the main finding. The SAEs studied here reconstruct well by conventional measures, yet they lose task information that PCA at equal error keeps, and their reconstruction carries only part of a confirmed causal effect at mid and late layers. Fine-tuning improved

task reconstruction and output faithfulness at layer 6, but no method met its pre-registered truth criterion.

PENDING: Revise after IMP-L, E1–E3 and M5; in particular, whether the conclusions hold for an externally trained SAE (E2).

13 Limitations and Threats to Validity

- **Small, data-limited SAEs.** 3072 latents (4×), trained on 101,600 document-start tokens for about 38 passes, not converged (validation-based checkpoint selection always chose the final step). The public `gpt2-small-res-jb` SAE used 300M tokens and 24,576 latents. Conclusions may not transfer.
- **One operating point.** L0 = 128 and width 3072 only.
- **Surrogate JumpReLU.** Not the published estimator; up to 85 % dead latents at layer 0.
- **Task-level sparsity mismatch.** Matching is on OWT; task prompts activate very different numbers of latents per architecture.
- **Weak behavioural signal.** The persona effect is 0.02–0.04 nats; the persona-meaning gate passed narrowly (0.701).
- **Seeds share data.** Seed spread understates uncertainty.
- **Test exposure.** The GoT test split was scored in earlier frozen-probe evaluations; the original RepE test split was touched in a smoke run and is retired with the confounded task.
- **Single model.** No SAE results on Gemma-2-2B [6]; Gemma Scope weights were not available on our machine.

14 Conclusion and Future Work

Across all layers of GPT-2 Small, three SAE architectures reconstruct activations well by conventional measures, but reconstruction quality predicts behavioural damage rather than information preservation, frozen-probe "truth loss" is largely perturbation sensitivity, SAEs never beat PCA at equal error on task information, and at mid and late layers the SAE reconstruction carries only about two-thirds of a confirmed causal effect. Future work: larger and longer-trained SAEs, the published JumpReLU estimator, task-level sparsity matching, and the same audit on Gemma-2-2B with Gemma Scope SAEs.

PENDING: Final conclusions after M5.

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A Project History and Corrections

- Layers were first trained with mixed optimiser procedures; all 108 SAEs were retrained with one uninterrupted 15k-step protocol (explained variance changed by at most 0.006; rankings unchanged).
- An architecture summary averaged unmatched runs; it was replaced by a ±10 % L0 matching rule.
- Earlier "SAE steering" never placed an SAE in the forward pass; it was reclassified.
- The truth-aware fine-tuning objective was circular (§10).
- The original RepE persona prompts were length-confounded; RepE-LM replaces them.
- Correlations pooled across layers were replaced by within-layer analyses (M0).
- Frozen-probe failures were re-analysed with noise and PCA references (M2).

Historical results (original RepE probes and steering, the layer-6 causal test with 200 controls, R1 honesty fine-tuning, R2 feature clamps, Gemma steering, Feature 974) are kept in the repository with their status labels and are not used as evidence here.

B Test-split Exposure

OWT test: used once, for the reconstruction table. GoT test: scored for all 108 SAEs with frozen probes and twice for historical layer-6 fine-tunes. RepE-LM test: never built or accessed. M0–M4 used no test rows.

C Reproducibility

Code, configurations, pre-registrations and result files: repository `Fall2026group4/fall-2026-group4`, branch `sae-research`. Software: Python 3.14, PyTorch 2.14.1, TransformerLens 3.9.0 [10], transformers 5.21.0, SAELens 6.54.12. Hardware: one NVIDIA A10G (23 GB). Tables and figures in this paper are generated by `scripts/shared/build_paper_artifacts.py` from the committed result files. SAE checkpoints (about 9 GB) are listed with SHA-256 hashes in `docs/checkpoint_registry.csv`.

PENDING: Command list per experiment; pipeline diagram (Review #8 F8: `.drawio` source and `.svg`).